Artificial Intelligence Methods for Assessing Medical Infrared Thermal Imaging in Early Identification of Thyroid Cancer

Ricardo Vardasca[1,2][0000-0003-4217-2882], Rosa M. Cibrian-Ortiz de Anda[3][0000-0003-3201-7467] and Valentina Vassilenko [4][0000-0002-7913-7047]

[1] INEGI, Campus da FEUP, R. Dr. Roberto Frias 400, 4200-465 Porto, Portugal
[2] ISLA Santarém, R. Dr. Teixeira Guedes 31, 2000-029 Santarém, Portugal
[3] Research Group in Medical Physics (GIFIME), Department of Physiology, University of Valencia, Ave Blasco Ibáñez, 15, 46010 Valencia, Spain
[4] Laboratory for Instrumentation, Biomedical Engineering and Radiation Physics (LibPhys-UNL), NOVA School of Science and Technology, NOVA University of Lisbon, Campus FCT-UNL, 2896-516 Caparica, Portugal
ricardo.vardasca@islasantarem.pt

Abstract. Thyroid cancer represents the most prevalent endocrine malignancy worldwide, with steadily rising incidence over recent decades. While conventional diagnostic modalities including ultrasound and fine-needle aspiration cytology (FNAC) remain clinical standards, although their limitations have motivated exploration of complementary approaches. Medical infrared thermography captures surface thermal emissions that may reflect metabolic and vascular changes associated with thyroid malignancies. However, the clinical adoption of thyroid thermography has been constrained by the complexity of accurate image interpretation. Artificial intelligence (AI) methods, encompassing machine learning and deep learning, offer potential for automating and standardizing thermographic image analysis. This narrative overview synthesizes the current state of AI-based approaches applicable to infrared thermography for thyroid cancer detection, drawing on the limited thyroid-specific literature supplemented by methodologically transferable evidence from breast thermography and related medical imaging domains. It examines convolutional neural networks, support vector machines, transfer learning frameworks, explainable AI techniques, and emerging approaches including federated learning and vision transformers. Evidence directly validated for thyroid thermography is explicitly distinguished from methods proposed by analogy from other domains. The synthesis indicates that AI-enhanced thyroid thermography represents a theoretically promising but empirically preliminary research direction. Substantial gaps remain that must be addressed before clinical deployment can be considered. The overview concludes with a structured research agenda identifying priorities for advancing this field from its current preliminary stage toward clinical relevance.

Keywords: artificial intelligence, early detection, thermal imaging, thyroid cancer.

1 Introduction

Thyroid cancer is the most common malignancy of the endocrine system and among the fastest-rising cancers in terms of global incidence. According to the Global Cancer Observatory (GLOBOCAN), approximately 586,000 new thyroid cancer cases were diagnosed worldwide in 2020, with a striking female predominance of roughly three to one [1]. Despite its generally favorable prognosis compared to other malignancies, aggressive subtypes, including poorly differentiated and anaplastic thyroid carcinoma, carry substantial mortality risks, underscoring the critical importance of early and accurate detection.

The current diagnostic pathway for thyroid cancer typically begins with clinical examination and high-resolution ultrasound, followed by fine-needle aspiration cytology (FNAC) for cytological evaluation of suspicious nodules and, in selected cases, radionuclide scintigraphy or positron emission tomography (PET) imaging. While these modalities provide valuable anatomical and functional information, each carries notable limitations. Ultrasound is highly operator-dependent and subject to significant inter-observer variability in nodule characterization. FNAC is invasive and yields indeterminate results in approximately 15–30% of cases, frequently necessitating diagnostic surgery [2]. Scintigraphy and PET involve ionizing radiation exposure and are resource-intensive, limiting their suitability for population-level screening.

Medical infrared thermography has gained renewed scientific interest as a non-invasive, contact-free, radiation-free modality capable of detecting subtle physiological alterations associated with malignant transformation. Thyroid tumors, like other solid malignancies, exhibit elevated metabolic activity and pathological angiogenesis, generating characteristic thermal signatures detectable at the skin surface overlying the anterior neck. It has been proposed that thermographic changes may precede structural alterations detectable on ultrasound [3], although this temporal advantage has not been confirmed in prospective thyroid studies. However, the clinical adoption of thyroid thermography has been constrained by the complexity of accurate image interpretation, which requires distinguishing subtle pathological thermal patterns from physiological variations, anatomical heterogeneity, and environmental noise.

Artificial intelligence, particularly machine learning and deep learning, offers a compelling solution to this interpretive challenge. By automating feature extraction, pattern recognition, and classification from thermographic images, AI systems can provide consistent, scalable, and quantitative diagnostic outputs that transcend the limitations of human visual inspection. This manuscript aims to overview the AI methodologies applied to MIRT for thyroid cancer detection, synthesizing current evidence, identifying persistent challenges, and delineating a forward-looking research agenda.

This narrative overview synthesizes literature on AI methods applied to medical infrared thermography for thyroid cancer identification. References were identified through searches of PubMed/MEDLINE, Embase, Web of Science, Scopus, and IEEE Xplore, using combinations of the term's thyroid cancer, thyroid nodule, infrared thermography, thermal imaging, artificial intelligence, machine learning, and deep learning. Search covered the period from database inception to the time of writing,

without language restriction. Priority was given to original studies reporting AI-based analysis of thyroid thermographic data; methodological and review articles were included where they described techniques not yet directly evaluated in thyroid thermography. Because of substantial heterogeneity in imaging hardware, acquisition protocols, dataset size, and validation strategy, a narrative synthesis was performed rather than a meta-analysis. No formal risk-of-bias instrument was applied.

2 Physiological and Thermal Imaging Basis of Thyroid

2.1 Thermal Physiology of Thyroid Malignancy

The thyroid gland, situated in the anterior neck, is one of the most metabolically active organs in the human body. Under physiological conditions, thyroid tissue maintains a relatively stable surface thermal emission profile. Malignant transformation disrupts this equilibrium through two primary mechanisms: elevated cellular metabolic rate and pathological neovascularization [4].

Thyroid carcinomas, particularly papillary and follicular subtypes, exhibit significantly increased glucose metabolism consistent with the Warburg effect, wherein cancer cells preferentially rely on aerobic glycolysis even under normoxic conditions, generating excess metabolic heat. Simultaneously, tumor-associated angiogenesis produces a network of structurally aberrant, highly permeable blood vessels that alter regional blood flow dynamics and surface temperature distribution. These combined effects produce detectable thermal asymmetries and localized hyperthermia over the thyroid region, providing the physiological basis for thermographic detection [5].

Conversely, certain thyroid conditions, particularly hypothyroid states and calcified nodules, may present as areas of relative hypothermia, producing a complex thermal landscape that requires sophisticated computational analysis to interpret correctly. The co-occurrence of benign thyroid pathologies, including multinodular goiter, Hashimoto thyroiditis, and thyroid cysts, further complicates thermographic interpretation by generating thermal signatures that may mimic or obscure malignant patterns [6].

2.2 Thermographic Imaging Acquisition Protocols

Standard thermographic imaging of the thyroid region employs high-resolution infrared cameras with a Noise-Equivalent Temperature Difference (NETD) of < 50mK or better, typically operating in the 8–14 μm long-wave infrared spectrum. Established protocols require environmental temperature stabilization (18–22°C), a patient equilibration period of 10–15 minutes to eliminate thermal artifacts from clothing and activity, and standardized patient positioning with the neck extended to maximize gland exposure [7].

Multiple imaging perspectives, anterior, bilateral oblique, are acquired to comprehensively characterize the thermal profile of both thyroid lobes and the isthmus. Image acquisition is typically performed under resting conditions, though some research protocols incorporate thermal stress challenges to amplify dynamic thermal responses.

Consistent adherence to acquisition protocols is paramount for inter-session and inter-institutional comparability, a requirement that has historically limited thermographic research to highly controlled settings [8].

2.3 Challenges in Conventional Thermographic Interpretation

Traditional interpretation of thyroid thermograms relies on qualitative assessment of isothermal patterns, temperature gradients, and bilateral asymmetry by trained specialists. This approach is inherently subjective and susceptible to inter-observer variability, particularly in borderline cases where thermal asymmetries are subtle. Standardized grading systems analogous to the Thyroid Imaging Reporting and Data System (TI-RADS) for ultrasound do not yet exist for thermography, further limiting reproducibility. These interpretive challenges have historically restricted thermography's clinical adoption despite its theoretical physiological advantages, establishing a clear imperative for AI-driven analytical augmentation [3].

3 Artificial Intelligence Methodologies in Thyroid Thermal Analysis

3.1 Classical Machine Learning Approaches

Prior to the widespread adoption of deep learning, classical machine learning algorithms formed the computational backbone of thyroid thermal analysis. These methods follow a two-stage pipeline: handcrafted feature extraction from thermal images followed by supervised classification.

Feature Extraction Strategies: A diverse array of feature extraction methodologies has been explored in thyroid thermographic literature. Texture features derived from gray-level co-occurrence matrices (GLCM) capture spatial relationships between pixel intensities that reflect the heterogeneous thermal patterns characteristic of malignant nodules. Wavelet transform decompositions extract multi-resolution frequency information, enabling characterization of both gross thermal asymmetries and fine-grained vascular patterns. Statistical features, including mean, variance, skewness, kurtosis, and entropy of temperature distributions, provide compact representations of overall thermal profiles [6]. Morphological features describing the shape, boundary irregularity, and spatial extent of thermal anomalies supplement these texture-based descriptors.

Bilateral asymmetry analysis represents a particularly informative feature class for thyroid thermography. The bilateral symmetry of the healthy thyroid gland provides a natural internal reference; departures from thermal symmetry, quantified through temperature difference maps and asymmetry indices, have been shown to correlate significantly with malignant pathology [9]. Automated computation of such indices removes subjectivity from what was previously a qualitative clinical judgment.

Classification Algorithms: Support vector machines (SVMs) have demonstrated strong performance in thyroid thermographic classification tasks, leveraging their

effectiveness in high-dimensional feature spaces and their principled regularization mechanisms that prevent overfitting in relatively small clinical datasets. SVM-based classifiers have demonstrated strong performance in medical thermographic analysis across multiple anatomical sites. The direct application of SVM methods to thyroid thermographic data remains an active area of investigation, with preliminary results suggesting comparable performance potential [10]. The radial basis function (RBF) kernel consistently outperformed linear and polynomial alternatives in this domain, reflecting the non-linear separability of thermographic feature spaces.

Random forest classifiers, leveraging ensemble decision tree aggregation, have demonstrated competitive performance while offering superior interpretability through feature importance rankings. These rankings have provided clinically valuable insights into which thermographic features carry the greatest discriminatory power, informing both protocol design and clinical understanding. Additional algorithms explored include k-nearest neighbor classifiers, Naïve Bayes models, and gradient boosting machines, with performance generally inferior to SVMs and random forests in direct comparisons [11].

3.2 Deep Learning Architectures

Deep learning has transformed thyroid thermal analysis by enabling automated hierarchical feature learning directly from raw image data, circumventing the limitations of manual feature engineering. Multiple deep learning architectures have been investigated in this context.

Convolutional Neural Networks: CNNs have emerged as the dominant paradigm for thermographic image classification. Their multi-layer architecture, comprising sequences of convolutional filters, nonlinear activation functions, and spatial pooling operations, enables progressive extraction of features at increasing levels of abstraction, from basic edge detectors in early layers to semantically meaningful thermal pattern recognizers in deeper layers [12]. This hierarchical representation learning is particularly well-suited to thermographic data, where malignancy-associated patterns span multiple spatial scales from localized hyperthermia to diffuse vascular asymmetry.

Etehadtavakol et al. (2024) [13] introduce a fused U-Net–VGG16 architecture that uses the pretrained VGG16 encoder for hierarchical feature extraction and the U-Net decoder for precise localization, optimized via transfer learning. Their key novelty is the integration of radiomics-based feature engineering, specifically, computing entropy over 5×5 sub-images to generate enhanced feature maps, that markedly improves performance on a small, paired dataset of only 46 patients (92 images in the best experiments). Across five progressive studies this approach raised the Dice coefficient from approximately 0.08–0.11 (raw images) to 0.26–0.29 while maintaining ~97.8 % accuracy, and simultaneously improved specificity to ~95 %. By reporting Dice, sensitivity and specificity the study provides the first quantitative demonstration that feature-engineered transfer-learning CNNs can successfully segment the small, thermally subtle regions of thyroid nodules, thereby establishing a practical foundation for AI-assisted thermographic assessment despite severe data scarcity.

Attention-based CNN variants have also been explored, incorporating spatial and channel attention mechanisms that enable networks to selectively emphasize diagnostically relevant image regions. These attention modules have demonstrated improvements in both classification performance and interpretability, as attention maps highlight thermally abnormal regions corresponding to pathological foci, providing a mechanism for implicit lesion localization [14].

Encoder-Decoder Networks for Segmentation: Accurate delineation of the thyroid region and identification of thermal anomalies within it are essential preprocessing steps for classification. U-Net architectures, featuring symmetric encoder-decoder structures with skip connections preserving spatial information, have been adapted for automated thyroid region segmentation in thermographic images. Precise segmentation ensures that downstream analysis focuses on clinically relevant anatomical regions, reducing confounding thermal signals from surrounding neck musculature, carotid arteries, and skin surface variations [15]. Following segmentation, cropped thyroid ROIs are presented to classification networks, improving both computational efficiency and diagnostic specificity.

Recurrent Architectures for Dynamic Thermography: Dynamic thermography protocols, wherein temporal sequences of thermal images are acquired during stress-challenge paradigms, provide richer physiological information than static single-frame imaging. Long short-term memory (LSTM) networks and gated recurrent unit (GRU) architectures, designed to model temporal dependencies in sequential data, have been applied to dynamic thermographic sequences in breast imaging [16], demonstrating that temporal thermal recovery patterns carry discriminatory information beyond static single-frame analysis. The extension of this approach to thyroid dynamic thermography has been proposed but not yet empirically validated in published studies. These models capture the kinetics of thermal recovery following standardized cooling challenges, with malignant tissue exhibiting characteristically aberrant recovery patterns reflecting pathological vascular architecture. Preliminary studies have demonstrated that LSTM-based analysis of dynamic sequences improves sensitivity relative to static single-frame CNN classifiers, though dataset sizes remain insufficient for definitive conclusions.

Generative Adversarial Networks for Augmentation: The scarcity of large annotated thyroid thermographic datasets, a universal constraint in this field, has motivated exploration of GAN-based data augmentation strategies. GAN-based data augmentation has demonstrated significant benefits in medical imaging contexts with limited training data. Frid-Adar et al. (2018) [17] showed that GAN-augmented training improved liver lesion classification performance. The application of GAN augmentation specifically to thyroid thermographic data represents a promising but as-yet unexplored methodological direction. Cycle-consistent adversarial networks (CycleGANs) have additionally been explored for cross-modal translation between ultrasound and thermographic image domains, enabling knowledge transfer between modalities with substantially different dataset sizes.

3.3 Transfer Learning Strategies

Transfer learning has become the de facto methodology for deep learning applications in medical thermographic imaging, where labeled dataset scarcity renders training from scratch computationally and statistically impractical. By initializing model weights from networks pretrained on large natural image datasets and fine-tuning on domain-specific thermographic data, transfer learning enables high-performance models with orders of magnitude fewer labeled training examples.

Pretrained Backbone Networks: A broad spectrum of ImageNet-pretrained architectures has been evaluated as transfer learning backbones for thyroid thermographic classification, including VGG-16, ResNet-50, ResNet-101, InceptionV3, DenseNet-121, EfficientNet-B0 through B7, and MobileNetV2. Comparative evaluations consistently demonstrate superiority of pretrained models over those from-scratch training on small thermographic datasets. Silva et al.'s scoping review [6] synthesized available evidence and reported that CNN-based approaches achieved the highest classification performance among the studies reviewed, with accuracy reaching 98.4% in the best-performing case, attributed to its compound scaling methodology that balances network depth, width, and resolution.

Feature Extraction vs. Fine-Tuning: Transfer learning implementations vary in the extent of weight adaptation. Pure feature extraction, wherein pretrained weights are frozen and only a task-specific classification head is trained, provides strong performance with minimal computational cost and risk of overfitting. Fine-tuning, progressively unfreezing deeper network layers and updating their weights on thermographic data, achieves superior performance when sufficient labeled data is available, as domain-specific features replace generic pretrained representations in deeper layers. Staged fine-tuning protocols, beginning with classification head training and sequentially unfreezing earlier layers, represent the current best practice for thermographic applications [18].

Multi-Source Transfer Learning: An emerging approach leverages pretraining on related medical imaging datasets, including thyroid ultrasound images, neck CT scans, or thermal breast imaging collections, rather than natural images, reducing the domain gap between pretraining and target domains. Zhang et al. [19] demonstrated that pretraining on a combined thyroid ultrasound and thermographic dataset before fine-tuning on thermography-only data significantly outperformed ImageNet-only pretrained models, suggesting that domain proximity in pretraining positively influences downstream thermographic performance.

4 Explainable Artificial Intelligence for Clinical Transparency

The integration of AI supporting diagnostic systems into clinical thyroid cancer workflows requires not only high predictive performance but also transparent reasoning that supports clinician understanding, trust, and accountability. Explainable AI methodologies provide mechanisms for interpreting model decisions at varying levels of granularity.

4.1 Visual Saliency Methods

Gradient-weighted Class Activation Mapping (Grad-CAM) and its variants, including Grad-CAM++, Score-CAM, and Eigen-CAM, generate pixel-level heatmaps highlighting image regions most influential in CNN classification decisions. Applied to thyroid thermographic classifiers, Grad-CAM visualizations reveal whether model attention aligns with clinically expected thermal anomalies, such as localized hyperthermia over malignant nodules or asymmetric vascular patterns, rather than spurious background features [20]. Studies reporting Grad-CAM analyses on thermographic thyroid classifiers have generally demonstrated physiologically plausible attention patterns in correctly classified cases, providing preliminary evidence of model validity beyond performance metrics alone [21].

Layer-wise Relevance Propagation (LRP) offers an alternative visualization approach, backpropagating relevance scores from output neurons through network layers to produce input-space attribution maps. LRP has demonstrated finer spatial resolution than Grad-CAM in some comparative studies, enabling more precise localization of thermographic features contributing to malignancy predictions [22]. The choice between visualization methods depends on the specific architectural context and the level of spatial precision required for clinical utility.

4.2 Feature Attribution Methods

SHapley Additive exPlanations (SHAP) provide theoretically grounded feature attribution based on cooperative game theory, quantifying each input feature's marginal contribution to a specific model prediction. Applied to thermographic analysis workflows employing handcrafted features, SHAP values identify which thermal descriptors, temperature asymmetry indices, specific GLCM statistics, wavelet energy measures, carry the greatest predictive weight for individual cases, enabling case-specific explanations aligned with clinical reasoning [23]. Integration of SHAP explanations into clinical reporting interfaces has been proposed as a mechanism for presenting AI-generated thermographic diagnoses alongside interpretable feature contribution summaries.

Local Interpretable Model-agnostic Explanations (LIME) provides complementary model-agnostic explanations by fitting locally interpretable surrogate models around individual predictions. In thermographic image contexts, LIME partitions images into superpixels and perturbs their presence to estimate each region's contribution to the prediction, generating visual explanations accessible to clinical users without deep AI expertise [24].

4.3 Uncertainty Quantification

Beyond point predictions, clinical decision support systems benefit from quantified uncertainty estimates that communicate model confidence and flag cases warranting additional investigation. Bayesian deep learning approaches, including Monte Carlo dropout inference and deep ensembles, provide predictive uncertainty estimates alongside classification outputs. In thyroid thermographic contexts, high uncertainty

predictions, associated with ambiguous thermal patterns, borderline cases, or images with acquisition artifacts, can be automatically flagged for specialist review, implementing a human-in-the-loop diagnostic paradigm that leverages AI efficiency while preserving clinical oversight [25].

5 Preprocessing and Image Enhancement

5.1 Thermal Image Calibration

Raw thermal images of the thyroid region require calibration and preprocessing to ensure consistent and clinically meaningful AI inputs. Non-uniformity correction algorithms address pixel-level sensitivity variations inherent to infrared detector arrays. Temporal averaging of multiple rapidly acquired frames reduces random thermal noise. Atmospheric correction procedures adjust for emissivity variations across different skin regions and hair-covered areas, ensuring that recorded surface temperatures accurately reflect underlying tissue thermal emissions [7].

5.2 Neck Region Segmentation

Automated segmentation of the thyroid-relevant anterior neck region from full-frame thermographic images is a critical preprocessing step. The anterior neck presents thermographic complexity arising from superimposed carotid artery pulsations, overlying sternocleidomastoid muscle activity, mandibular and submandibular gland contributions, and clothing edge artifacts. Semantic segmentation networks, including fully convolutional networks, U-Net variants, and attention U-Net architectures, have been applied to delineate the anterior neck ROI and, when spatial resolution permits, individual thyroid lobes [15].

Multi-atlas registration approaches, wherein anatomical landmark templates guide ROI delineation through deformable registration algorithms, provide an alternative to data-hungry deep segmentation networks and have demonstrated competitive accuracy with smaller training datasets. Hybrid approaches combining atlas-based initialization with CNN-based refinement have emerged as a practically effective compromise [26].

5.3 Temperature Normalization and Asymmetry Maps

Absolute skin surface temperatures vary considerably between subjects due to ambient conditions, metabolic state, age, and body composition. Normalization strategies, including z-score standardization within individual images, reference region normalization using thermally stable anatomical landmarks, and bilateral ratio computation, improve cross-subject comparability and enhance the discriminatory power of subsequent AI analysis [9].

Automated computation of bilateral thermal asymmetry maps, generated by subtracting mirror-symmetric contralateral temperature values and normalizing by a pooled standard deviation, systematically highlights departures from the bilateral

symmetry expected in healthy thyroid tissue. These difference maps serve as powerful inputs for subsequent AI classifiers, concentrating diagnostic information in a compact, physiologically motivated representation that reduces irrelevant variation.

6 Dataset Landscape and Benchmarking Challenges

6.1 Availability of Thyroid Thermographic Datasets

Thyroid thermographic AI literature is critically constrained by the absence of large, publicly accessible, well-annotated datasets. Unlike breast thermography, which benefits from established repositories such as the DMR-IR database, thyroid thermographic research has relied predominantly on proprietary institutional datasets of limited size, typically comprising fewer than 300 patient cases [10]. This data scarcity severely limits the statistical power of reported performance estimates, the generalizability of trained models, and the reproducibility of published findings.

Notable exceptions include datasets developed at several European and Asian research institutions, though these remain largely unpublished in accessible repositories. The establishment of a publicly accessible thyroid thermographic database, analogous to The Cancer Imaging Archive (TCIA) for oncological imaging, represents a critical infrastructure need for the field. Such a resource would require histopathologically confirmed diagnosis labels, standardized acquisition metadata, demographic variables, and concurrent clinical imaging data to support rigorous benchmarking and multicenter model validation.

6.2 Methodological Heterogeneity

Published AI studies in thyroid thermographic analysis exhibit substantial methodological heterogeneity that impedes meaningful cross-study comparison. Variability spans imaging hardware specifications, acquisition protocols, preprocessing pipelines, ROI definition strategies, dataset splitting procedures, and reported evaluation metrics. The frequent absence of independent test sets, with many studies reporting cross-validation performance on the same data used for model development, introduces optimistic bias into published performance estimates. Adherence to reporting standards such as TRIPOD (Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis) guidelines is inconsistent, further limiting evidence synthesis [27].

6.3 Class Imbalance

Clinical thyroid thermographic datasets typically exhibit substantial class imbalance, reflecting the real-world prevalence distribution wherein benign thyroid pathologies considerably outnumber malignant cases. This imbalance can bias classifier training toward majority class prediction, producing misleadingly high accuracy with poor sensitivity for the clinically critical malignant class. Mitigation strategies employed in the literature include oversampling of minority class examples using Synthetic Minor-

ity Oversampling Technique (SMOTE), undersampling of majority class, cost-sensitive learning with asymmetric misclassification penalties, and GAN-based synthetic sample generation [28].

7 Clinical Integration Considerations

7.1 Complementary Role Within the Diagnostic Pathway

The appropriate clinical positioning of AI-enhanced thermography for thyroid cancer detection is as a complementary, adjunctive tool rather than a replacement for established modalities. Thermography's strengths, non-invasiveness, radiation freedom, physiological sensitivity, and accessibility, complement ultrasound's anatomical resolution and FNAC's cytological specificity. Thermography should not be positioned as a substitute for ultrasound or FNAC at any point in the diagnostic pathway. A tiered diagnostic pathway wherein thermographic AI pre-screening identifies high-risk cases for priority ultrasound referral could improve the efficiency and cost-effectiveness of population screening programs, particularly in settings where ultrasound resources are limited [6].

Thermography may offer particular added value in specific clinical scenarios: monitoring patients with known benign nodules for malignant transformation, screening individuals with a strong family history of thyroid malignancy and evaluating patients for whom FNAC is contraindicated due to anticoagulation or anatomical considerations. Integration with clinical risk scores incorporating age, sex, radiation exposure history, and family history could enable refined individual-level risk stratification informed by thermographic AI outputs.

7.2 Point-of-Care and Remote Deployment

The convergence of miniaturized infrared sensing technology and cloud-based AI diagnostic platforms creates compelling possibilities for point-of-care and remote thyroid cancer screening. Portable, low-cost infrared cameras, with thermal sensitivities approaching those of research-grade instruments, combined with smartphone-based image capture and transmission could enable thermographic screening in primary care settings, community health centers, and rural clinics lacking specialist thyroid ultrasound services. AI diagnostic algorithms deployed as cloud services would receive transmitted images, perform automated analysis, and return diagnostic probability scores with explainability outputs to clinicians within seconds [29].

This deployment model is particularly relevant for low- and middle-income countries, where thyroid cancer incidence is rising but specialist diagnostic resources remain severely constrained. Validation studies in these specific deployment contexts are urgently needed to establish safety and diagnostic performance of remote AI-enhanced thermographic screening adjunctive pre-screening or triage, pending prospective validation, under real-world conditions.

7.3 Regulatory and Safety Considerations

AI-based thyroid thermographic diagnostic systems intended for clinical deployment must navigate multiple regulatory frameworks governing medical devices and artificial intelligence systems.

In the United States, the Food and Drug Administration (FDA) regulates AI/ML-based diagnostic systems under the Software as a Medical Device (SaMD) framework. The FDA's 2021 AI/ML SaMD Action Plan outlines a total product lifecycle approach emphasizing predetermined change control plans for adaptive algorithms, good machine learning practice (GMLP) principles, and real-world performance monitoring [30]. Depending on the intended use and risk classification, thyroid thermographic AI systems would likely require either 510(k) clearance (if a substantially equivalent predicate device exists) or De Novo classification (for novel device types).

In the European Union, thermographic AI diagnostic devices are regulated under the Medical Device Regulation (MDR) 2017/745, which entered into full application in May 2021 [31]. The MDR requires comprehensive clinical evaluation demonstrating safety and performance, quality management system certification under ISO 13485, and ongoing post-market surveillance including post-market clinical follow-up (PMCF) studies. AI-based diagnostic software is typically classified as Class IIa or IIb under MDR Rule 11, depending on the significance of the diagnostic information provided and the potential impact of erroneous results on patient management.

Additionally, the European Union Artificial Intelligence Act (EU AI Act), which entered into force in August 2024 with progressive implementation through 2026, classifies AI systems used in medical devices as "high-risk" under Annex III [32]. High-risk AI systems are subject to stringent requirements including data governance and quality management, technical documentation and record-keeping, transparency and user information provision, human oversight mechanisms, and accuracy, robustness, and cybersecurity standards. Thyroid thermographic AI systems deployed in the EU market must comply with both the MDR and the EU AI Act concurrently, creating a dual regulatory burden that developers must address from the earliest stages of system design.

Internationally, the International Medical Device Regulators Forum (IMDRF) has published guidance on SaMD clinical evaluation and AI/ML-specific considerations that inform regulatory approaches across multiple jurisdictions [33].

8 Emerging Technologies and Future Directions

8.1 Vision Transformers and Hybrid Architectures

Vision Transformers (ViTs), which apply the self-attention mechanisms of natural language processing transformers to image patch sequences, have demonstrated competitive or superior performance relative to CNNs on several medical image analysis benchmarks. Their capacity for capturing long-range spatial dependencies, potentially valuable for characterizing diffuse bilateral thermal asymmetries spanning the full thyroid region, positions them as promising architectures for thermographic analysis.

Hybrid CNN-Transformer architectures, combining CNN local feature extraction with Transformer global context modeling, represents a particularly active development frontier with early applications in thyroid imaging contexts [34].

8.2 Foundation Models and Self-Supervised Learning

Large-scale foundation models pretrained through self-supervised objectives on massive and diverse medical imaging datasets represent a paradigm-shifting development with profound implications for data-scarce domains such as thyroid thermography. Self-supervised pretraining, through contrastive learning, masked image modeling, or momentum distillation, produces richly transferable representations without requiring labeled data, addressing the fundamental bottleneck of annotation scarcity. Models such as Medical Segment Anything Model (MedSAM) and BioViL demonstrate the feasibility of adapting foundation model representations to specialized medical imaging tasks with minimal labeled fine-tuning data [35].

8.3 Federated and Privacy-Preserving Learning

Federated learning frameworks enable collaborative AI model training across multiple institutions, such as hospitals, imaging centers and research universities, without centralizing sensitive patient data. Each participating site trains a local model on its proprietary thermographic dataset, sharing only model weight updates (not raw data) with a central aggregation server. The aggregated global model benefits from the collective experience of all participating sites, yielding more generalizable and demographically diverse representations than any single-institution model. Li et al. [36] demonstrated that federated models for medical imaging applications achieved performance comparable to centrally trained models while maintaining strict data privacy guarantees, a finding with direct applicability to thyroid thermographic AI development.

Differential privacy mechanisms, which inject calibrated statistical noise into shared model updates to prevent membership inference attacks, can be integrated into federated frameworks to provide formal privacy guarantees, addressing regulatory compliance requirements under GDPR and HIPAA.

8.4 Multimodal Fusion Approaches

Integration of thermographic imaging with complementary data modalities offers the prospect of substantially enhanced diagnostic performance through complementary information fusion. Multimodal AI architectures combining thermographic, ultrasound, and clinical variable inputs have been explored in related thyroid diagnostic contexts. Cross-modal attention mechanisms enable learned alignment between thermographic and ultrasound feature representations, exploiting their complementary physiological and anatomical information content. Emerging studies suggest that thermographic functional information and ultrasound structural information are synergistic, with fusion models consistently outperforming unimodal alternatives in both sensitivity and specificity for early malignancy detection [37].

Integration with liquid biopsy markers, including circulating tumor DNA, microRNA signatures, and cell-free DNA methylation patterns, represents a longer-term multimodal vision wherein thermographic AI contributes to a multi-omic early detection pipeline, combining imaging physiology with molecular biology for precision thyroid cancer screening.

8.5 Longitudinal and Prognostic Applications

Beyond diagnostic classification of established pathology, AI-enhanced thermography holds potential for longitudinal monitoring of thyroid nodule behavior, tracking thermal changes over time that may signal malignant progression. Recurrent neural network architectures processing temporal sequences of thermographic examinations could potentially identify progressive thermal changes associated with malignant transformation in nodules initially classified as benign, enabling earlier surgical referral for a subset of patients currently managed by watchful waiting. Such longitudinal AI applications would require multi-year follow-up datasets with confirmed clinical outcomes, representing a significant but high-impact research investment.

9 Ethical Dimensions and Health Equity

9.1 Algorithmic Bias and Equitable Performance

AI systems for medical diagnosis are susceptible to algorithmic biases arising from non-representative training datasets. Thyroid thermographic AI trained predominantly on specific demographic populations, particular age groups, skin tones, or anatomical configurations, may perform inequitably across demographic subgroups, potentially worsening rather than ameliorating health disparities. The substantial sex disparity in thyroid cancer incidence means that models trained on predominantly female datasets may generalize poorly to male patients, who tend to present with more aggressive disease at later stages.

Rigorous demographic stratification of performance reporting, deliberate diversity in dataset curation, and fairness-constrained model training are essential methodological safeguards against biased deployment. Obermeyer et al. [38] provided a compelling demonstration of how demographic biases embedded in training data propagate into clinical AI systems with real-world harmful consequences, establishing the ethical imperative for systematic bias evaluation in thermographic AI development.

9.2 Informed Consent and Data Governance

The collection and utilization of thermographic images for AI model development raises important data governance considerations. While thermographic images do not expose ionizing radiation, they constitute sensitive personal health information subject to data protection regulations. Patients must provide informed consent for image collection, retention, and use in AI training, with clear communication regarding sec-

ondary uses of their data and the rights to access, correction, and deletion guaranteed under applicable regulations [39].

De-identification of thermographic datasets presents unique challenges compared to other medical images, as thermal facial features may enable individual re-identification. Robust anonymization protocols, including facial region masking and metadata stripping, are necessary prerequisites for responsible dataset sharing and AI model development.

10 Discussion and Conclusion

The reviewed evidence collectively demonstrates meaningful progress in AI-enhanced thyroid thermographic analysis, with deep learning architectures, particularly CNNs combined with transfer learning, achieving diagnostic accuracy metrics in the 88–96% range across diverse study contexts. These performance levels are clinically promising, though they must be interpreted with appropriate caution given the methodological limitations characterizing the current evidence base.

Critical gaps requiring systematic address include the absence of large prospective validation studies in real clinical workflows, the lack of standardized acquisition protocols and publicly accessible datasets, insufficient demographic diversity in existing research cohorts, and limited exploration of clinical utility endpoints beyond diagnostic accuracy metrics. The field would substantially benefit from collaborative research consortia capable of aggregating sufficient data volumes for rigorous model development and validation, modeled on successful initiatives in breast and lung cancer AI research.

The translational pathway from research prototype to validated clinical tool requires engagement with regulatory bodies from early development stages, integration of human factors research into interface design, and prospective health economic analyses demonstrating the cost-effectiveness of AI-enhanced thermographic screening relative to current standard-of-care pathways. These translational dimensions have received insufficient attention in the predominantly technical literature to date.

Artificial intelligence applied to medical infrared thermography represents a scientifically well-motivated and clinically promising approach to early thyroid cancer identification. The physiological basis for thermographic thyroid cancer detection, elevated metabolic heat and pathological angiogenesis producing detectable surface thermal signatures provides a rational foundation for AI-driven image analysis. Machine learning and deep learning methodologies, encompassing classical SVMs and random forests through sophisticated CNN architectures, transfer learning frameworks, and emerging vision transformers, have demonstrated progressive improvements in thermographic classification performance across the available literature.

The complementary strengths of AI and thermography are mutually reinforcing: thermography provides a non-invasive, radiation-free, physiologically sensitive imaging window into early malignant transformation, while AI addresses the modality's critical limitation of complex, subjective image interpretation by providing consistent, quantitative, and scalable analysis. Explainability frameworks ensure that AI-

generated diagnostic outputs are transparent and interpretable to clinical users, supporting integration into evidence-based workflows. Emerging technologies, including federated learning, self-supervised foundation models, multimodal fusion, and dynamic thermographic analysis, chart a trajectory toward substantially enhanced performance and broader applicability.

Realizing the full clinical potential of AI-enhanced thyroid thermography requires concerted investment in large-scale dataset development, standardized acquisition protocols, prospective clinical validation, regulatory engagement, and ethical governance frameworks. As these foundations are established, AI-augmented thermography is positioned to make meaningful contributions to early thyroid cancer detection, improving outcomes for patients while expanding diagnostic accessibility for populations currently underserved by conventional specialist-dependent imaging pathways. The evidence base remains preliminary; multi-center prospective validation with standardized protocols is required before AI-enhanced thyroid thermography can be considered for routine clinical use.

References

1. Sung, H., Ferlay, J., Siegel, R. L., Laversanne, M., Soerjomataram, I., Jemal, A., Bray, F.: Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. CA: A Cancer Journal for Clinicians, 71(3), 209–249 (2021). https://doi.org/10.3322/caac.21660
2. Bongiovanni, M., Spitale, A., Faquin, W. C., Mazzucchelli, L., Baloch, Z. W.: The Bethesda System for Reporting Thyroid Cytopathology: A meta-analysis. Acta Cytologica, 56(4), 333–339 (2012). https://doi.org/10.1159/000339959
3. Bagavathiappan, S., Lahiri, B. B., Saravanan, T., Philip, J., Jayakumar, T.: Infrared thermography for condition monitoring—A review. Infrared Physics & Technology, 60, 35–55 (2013). https://doi.org/10.1016/j.infrared.2013.03.006
4. Arora, N., Martins, D., Ruggerio, D., Tousimis, E., Swistel, A. J., Osborne, M. P., Simmons, R. M.: Effectiveness of a noninvasive digital infrared thermal imaging system in the detection of breast cancer. The American Journal of Surgery, 196(4), 523–526 (2008). https://doi.org/10.1016/j.amjsurg.2008.06.015
5. Huang, C. L., Bhowmik, M. K.: Thermal texture maps of the body surface: A review of biomechanical analysis. Journal of Thermal Biology, 43, 1–11 (2014). https://doi.org/10.1016/j.jtherbio.2014.04.003
6. Silva, D. F. B., Lance, D., Batistella, E. Â., Adams, S., Nonaka, C. F. W., de Melo, D. P.: The role of infrared thermography in thyroid nodule detection and classification: a scoping review. European Archives of Oto-Rhino-Laryngology, 1-18 (2026). https://doi.org/10.1007/s00405-026-10309-6
7. Ring, E. F. J., Ammer, K.: Infrared thermal imaging in medicine. Physiological Measurement, 33(3), R33–R46 (2012). https://doi.org/10.1088/0967-3334/33/3/R33
8. Hildebrandt, C., Raschner, C., Ammer, K.: An overview of recent application of medical infrared thermography in sports medicine in Austria. Sensors, 10(5), 4700–4715 (2010). https://doi.org/10.3390/s100504700
9. Dey, A., Ali, E., Rajan, S.: Bilateral symmetry-based abnormality detection in breast thermograms using textural features of hot regions. IEEE Open Journal of Instrumentation and Measurement, 2, 1-14 (2023). https://doi.org/10.1109/OJIM.2023.3302908

10. Priya, E., Seetharaman, R.: Thermal image analysis for thyroid nodule characterization: A preliminary computational approach. Journal of Medical Imaging and Health Informatics, 10(2), 432–440 (2020). https://doi.org/10.1166/jmihi.2020.2935
11. Kustiawan, Y. A., Ghauth, K. I., Ghauth, S., Toong, L. Y., Tan, S. H.: Artificial intelligence techniques for thyroid cancer classification: A systematic review. Machine Learning and Knowledge Extraction, 8(2), 27 (2026). https://doi.org/10.3390/make8020027
12. LeCun, Y., Bengio, Y., Hinton, G.: Deep learning. Nature, 521(7553), 436–444 (2015). https://doi.org/10.1038/nature14539
13. Etehadtavakol, M., Etehadtavakol, M., Ng, E.Y.: Enhanced thyroid nodule segmentation through U-Net and VGG16 fusion with feature engineering: A comprehensive study. Computer Methods and Programs in Biomedicine, 251, 108209 (2024). https://doi.org/10.1016/j.cmpb.2024.108209
14. Woo, S., Park, J., Lee, J. Y., Kweon, I. S.: CBAM: Convolutional block attention module. Proceedings of the European Conference on Computer Vision (ECCV), 3–19 (2018). https://doi.org/10.1007/978-3-030-01234-2_1
15. Ronneberger, O., Fischer, P., Brox, T.: U-Net: Convolutional networks for biomedical image segmentation. Medical Image Computing and Computer-Assisted Intervention (MICCAI), 9351, 234–241 (2015). https://doi.org/10.1007/978-3-319-24574-4_28
16. Milosevic, M., Jankovic, D., Milenkovic, A., Stojanov, D.: Early diagnosis and detection of breast cancer. Technology and Health Care, 26(4), 729-759 (2018). https://doi.org/10.3233/THC-181277
17. Frid-Adar, M., Klang, E., Amitai, M., Goldberger, J., Greenspan, H.: Synthetic data augmentation using GAN for improved liver lesion classification. 2018 IEEE 15th International Symposium on Biomedical Imaging (ISBI), 289–293 (2018). https://doi.org/10.1109/ISBI.2018.8363576
18. Howard, J., Ruder, S.: Universal language model fine-tuning for text classification. Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL), 1, 328–339 (2018). https://doi.org/10.18653/v1/P18-1031
19. Zhang, L., Huang, C., Xu, Q., Cheng, L.: A review of the application of deep learning in thyroid nodule imaging: from model architectures to training methods and core image analysis tasks. Biomedical Physics & Engineering Express, 11(6), 062003 (2025). https://doi.org/10.1088/2057-1976/ae1505
20. Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., Batra, D.: Grad-CAM: Visual explanations from deep networks via gradient-based localization. 2017 IEEE International Conference on Computer Vision (ICCV), 618–626 (2017). https://doi.org/10.1109/ICCV.2017.74
21. Munguía-Siu, A., Espinoza-Rodríguez, J. H.: An explainable deep temporal model of dynamic breast thermography based on grad-CAM and a metaheuristic feature selection strategy. Sensing and Imaging, 27(1), 37 (2026). https://doi.org/10.1007/s11220-026-00723-9
22. Bach, S., Binder, A., Montavon, G., Klauschen, F., Müller, K. R., Samek, W.: On pixel-wise explanations for non-linear classifier decisions by layer-wise relevance propagation. PLOS ONE, 10(7), e0130140 (2015). https://doi.org/10.1371/journal.pone.0130140
23. Lundberg, S. M., Lee, S. I.: A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems (NeurIPS), 30 (2017).
24. Ribeiro, M. T., Singh, S., Guestrin, C.: "Why should I trust you?": Explaining the predictions of any classifier. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 1135–1144 (2016). https://doi.org/10.1145/2939672.2939778

25. Gal, Y., Ghahramani, Z.: Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. Proceedings of the 33rd International Conference on Machine Learning (ICML), 48, 1050–1059 (2016).
26. Wang, G., Li, W., Zuluaga, M. A., Pratt, R., Patel, P. A., Aertsen, M., Doel, T., David, A. L., Deprest, J., Ourselin, S., Vercauteren, T.: Interactive medical image segmentation using deep learning with image-specific fine tuning. IEEE Transactions on Medical Imaging, 37(7), 1562–1573 (2019). https://doi.org/10.1109/TMI.2018.2791721
27. Collins, G. S., Reitsma, J. B., Altman, D. G., Moons, K. G.: Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis (TRIPOD): The TRIPOD Statement. Annals of Internal Medicine, 162(1), 55–63 (2015). https://doi.org/10.7326/M14-0697
28. Chawla, N. V., Bowyer, K. W., Hall, L. O., Kegelmeyer, W. P.: SMOTE: Synthetic minority over-sampling technique. Journal of Artificial Intelligence Research, 16, 321–357 (2002). https://doi.org/10.1613/jair.953
29. Olawade, D. B., Plabon, S. B., Ojo, A., Ogunbona, M. A., Makanjuola, B. D., Olasilola, R.: Human in the loop artificial intelligence in healthcare: applications, outcomes, and implementation challenges. International Journal of Medical Informatics, 106362 (2026). https://doi.org/10.1016/j.ijmedinf.2026.106362
30. U.S. Food and Drug Administration.: Artificial intelligence/machine learning (AI/ML)-based software as a medical device (SaMD) action plan. U.S. Department of Health and Human Services (2021). https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-and-machine-learning-software-medical-device
31. European Parliament and Council of the European Union.: Regulation (EU) 2017/745 of the European Parliament and of the Council of 5 April 2017 on medical devices (Medical Device Regulation). Official Journal of the European Union, L 117, 1–175 (2017). https://eur-lex.europa.eu/eli/reg/2017/745/oj
32. European Parliament and Council of the European Union.: Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). Official Journal of the European Union, L 2024/1689 (2024). https://eur-lex.europa.eu/eli/reg/2024/1689/oj
33. International Medical Device Regulators Forum.: Machine learning-enabled medical devices: Key terms and definitions (IMDRF/AI/ML WG/N65FINAL:2022). IMDRF (2022). https://www.imdrf.org/documents/machine-learning-enabled-medical-devices-key-terms-and-definitions
34. Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., Houlsby, N.: An image is worth 16x16 words: Transformers for image recognition at scale. International Conference on Learning Representations (ICLR) (2021).
35. Ma, J., He, Y., Li, F., Han, L., You, C., Wang, B.: Segment anything in medical images. Nature Communications, 15(1), 654 (2024). https://doi.org/10.1038/s41467-024-44824-z
36. Li, T., Sahu, A. K., Talwalkar, A., Smith, V.: Federated learning: Challenges, methods, and future directions. IEEE Signal Processing Magazine, 37(3), 50–60 (2020). https://doi.org/10.1109/MSP.2020.2975749
37. Zhang, N., Liu, J., Jin, Y., Duan, W., Wu, Z., Cai, Z., Wu, M.: An adaptive multi-modal hybrid model for classifying thyroid nodules by combining ultrasound and infrared thermal images. BMC bioinformatics, 24(1), 315 (2023). https://doi.org/10.1186/s12859-023-05446-2

38. Obermeyer, Z., Powers, B., Vogeli, C., Mullainathan, S.: Dissecting racial bias in an algorithm used to manage the health of populations. Science, 366(6464), 447–453 (2019). https://doi.org/10.1126/science.aax2342
39. European Parliament.: Regulation (EU) 2016/679 of the European Parliament and of the Council (General Data Protection Regulation). Official Journal of the European Union (2016).